



MINING DEALER BEST PRACTICE

Chassis Stands Guidelines for Effective Assembly on 794AC Electric Mining Trucks

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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July 2023
1.10-220728-1

1.0 Introduction

During the NPI (New Product Introduction) process of the 794AC truck in Brazil, one of the challenges encountered was the lack of clear guidelines regarding the specifications and positioning of Chassis Stands for use as support points for the truck during the assembly process.

The Customer safety guidelines recommend the positioning of the chassis must be done on fixed Stands and without the use of wood parts, thus requiring the manufacture of Chassis Stands in the correct size for each stage of the assembly process.

After some information received from the experiences of other Dealers and some initial studies, the Chassis Stands for the trucks were made and utilized on the first assembly of 794AC Trucks.

Some Chassis Stands size and placement details needed to be changed and adjusted during the first assembly. Based on the experience acquired, it is possible to determine the specifications of the Chassis Stands, which were used and validated in a new assembly at the same client, but at another site/mine.

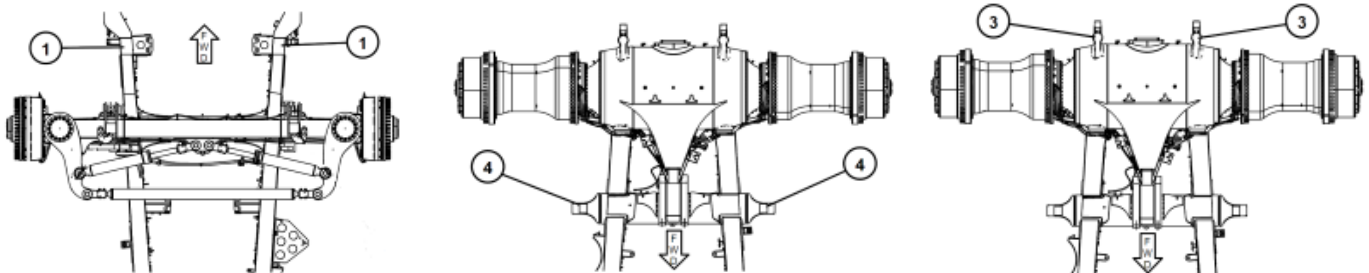
Based on the difficulties encountered and because it is a critical item related to cargo handling, the objective of this Best Practice is to present the sequencing of use and specifications of the Chassis Stands, to use as information reference about assembly.

2.0 Best Practice Description

With the introduction of new equipment by the engineering team, the planning of each stage of the execution and assembly process needed to be defined. The teams did not have previous experience in the assembly of 794AC model.

We searched SIS2.0 for information and recommendations for the Chassis Stands (height, dimensions of the upper base and other specifications) and we did not find them.

We also research Jacking Locations (SEBU9099-08) on the dimensions of the mainframe support surfaces and did not find the information easily. Items illustrated below (1), (2) and (3).



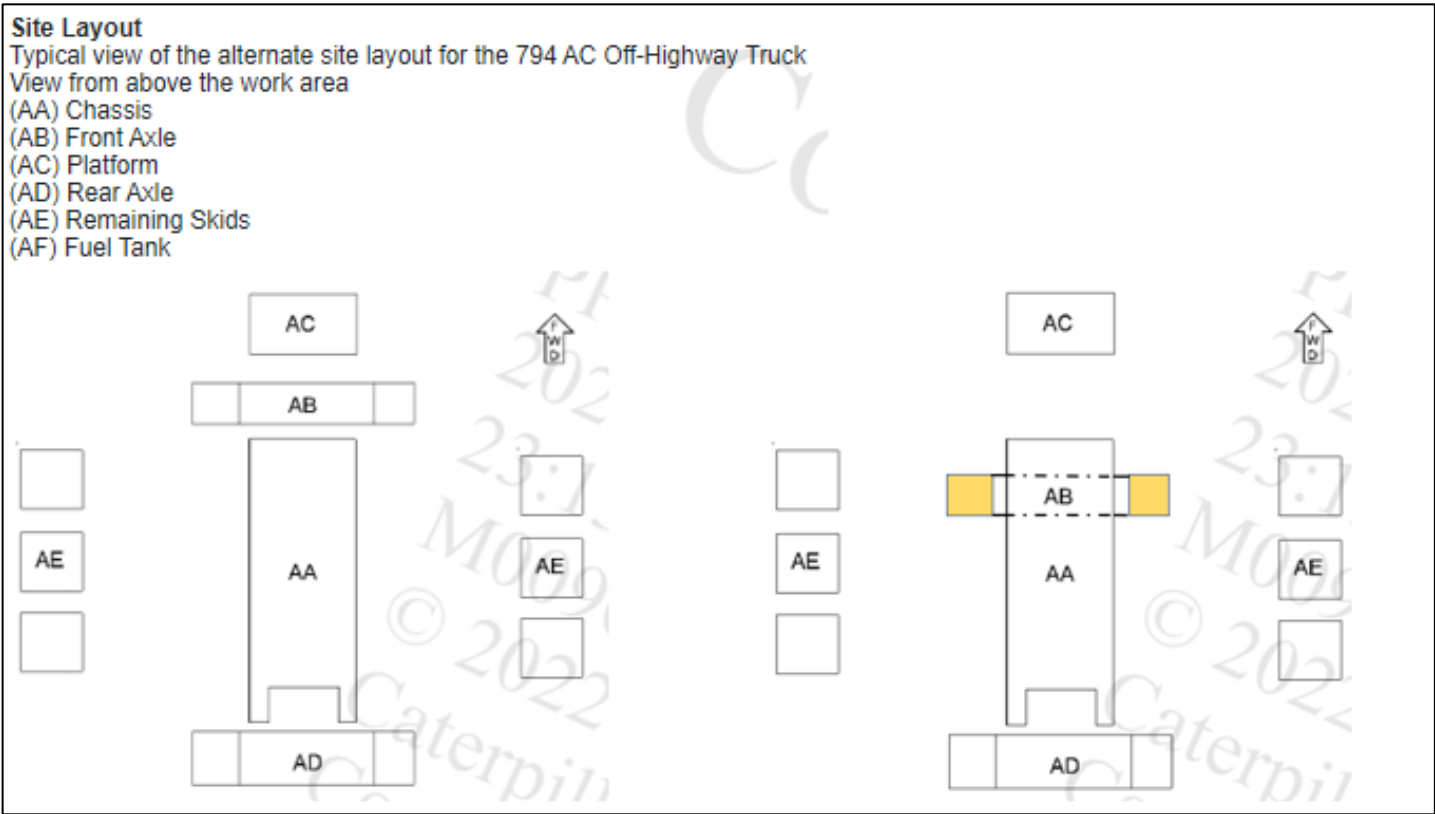
With the information provided from other dealers, about the assembly process, we got the dimensions of the main chassis support bases and some examples of Chassis Stands. But these models were not in line with our goal of prioritizing safety and productivity during the assembly process.

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The objective was to work with the mainframe as close to the ground as possible and on the front axle, to simplify the technicians work and minimize risks according to Customer Safety guidelines.

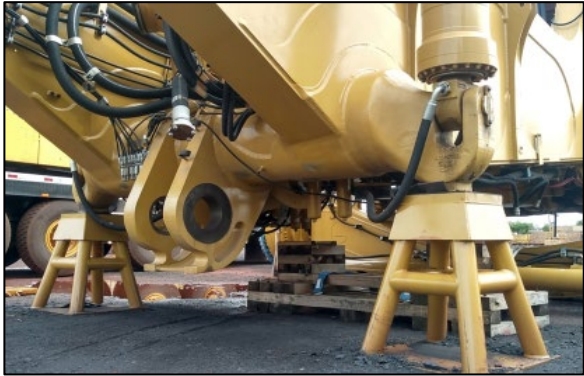
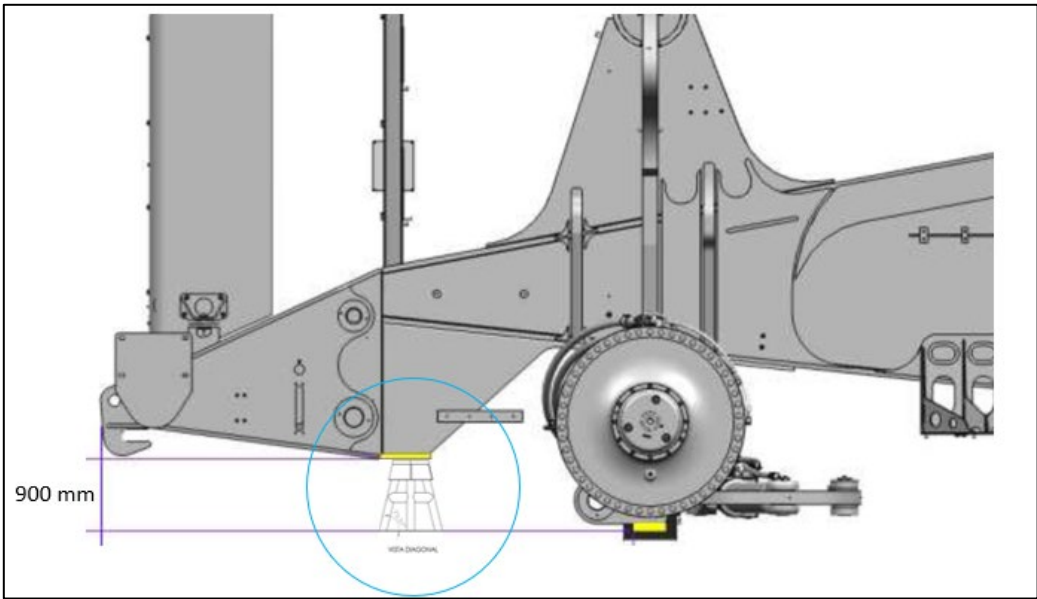
Unlike the assembly procedure (M0090261) which mentions the movement of the main frame about the front axle to install the fixing links, we choose to leave the main chassis static on the Chassis Stands and move the axle front about the main chassis during the link installation step. (Safety Reasons)

With the Activity, we managed to minimize the risks and exposure of our technicians to the load suspended by cranes.



Site Layout

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An important factor was to keep the chassis close to ground level during the assembly of the components, thus increasing the safety of the technicians. We positioned the main chassis on 900 mm stands, ensuring that the front axle, rear axle, and other components could be moved and installed without difficulty.

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After installing the **Front Axle**, we use a **500 mm** support only to support the center of the front axle.



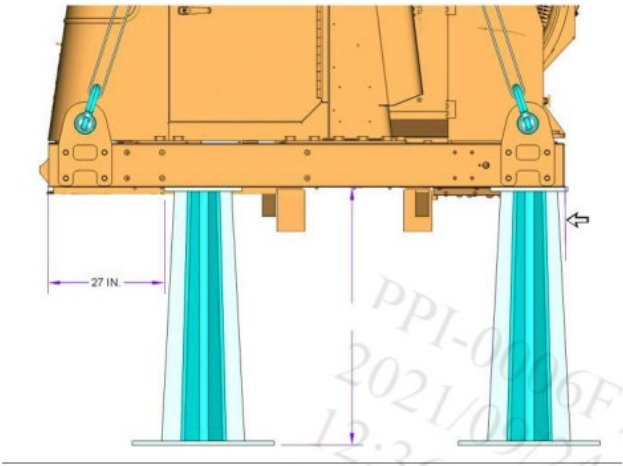
After installing the Rear Axle, we use 700 mm Chassis Stands. To support the Axle at a height compatible with the installation of the rear suspensions. Thus, avoiding unnecessary load movements and the use of support equipment reduces costs and increases the safety of the activity.



Superstructure positioned on **1600 mm Chassis Stands**.
Attention to the upper base: **Measures 2"x15"x11"**.

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Instructions for positioning the Chassis Stands can be found in the assembly instruction: **Assembly Procedure for the Off-Highway Truck - 794AC (M0090261)**



Our strategy was to work installing as many parts and components as possible with the truck supported by four **900 mm** Chassis Stands height and load capacity of **80 Metric Tons**.

After installing the superstructure, we leave the equipment ready for the installation of the front and rear wheels. With the support of two cranes, we lift the equipment and position it on the **1600 mm** Chassis Stands, thus providing the standard height for the tire handler to install the equipment wheels and tires.

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NOTE:

Before lifting the equipment to check on the **1600 mm** Chassis Stands, it is necessary to apply oil and nitrogen to the suspensions. The Chassis Stands are positioned under the **Front Axle** support base and Rear Axle base: **Servicing the Suspension Cylinders for 794 AC, 796 AC, and 798 AC Off-Highway Trucks - (M0092742-05).**

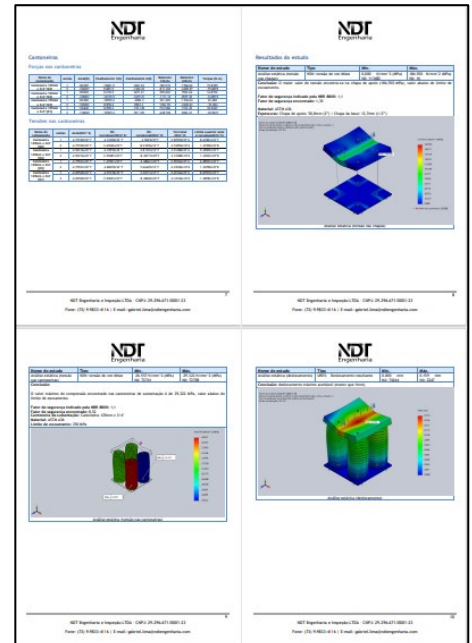
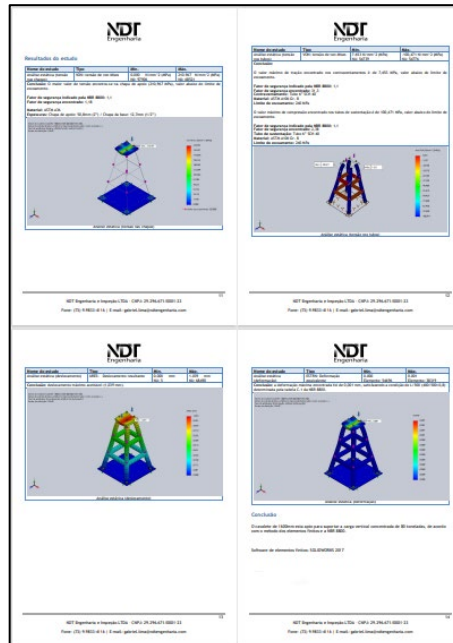
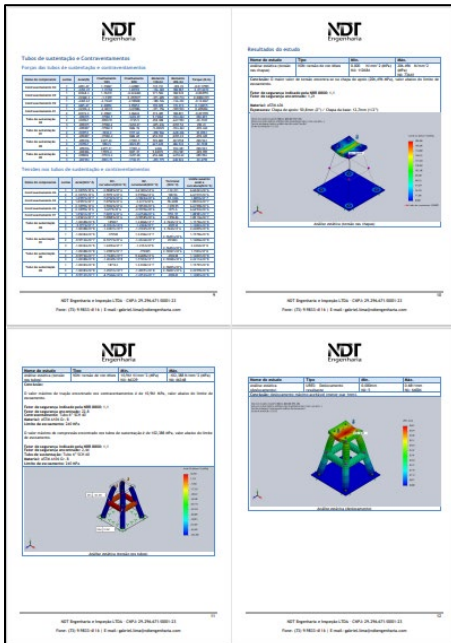
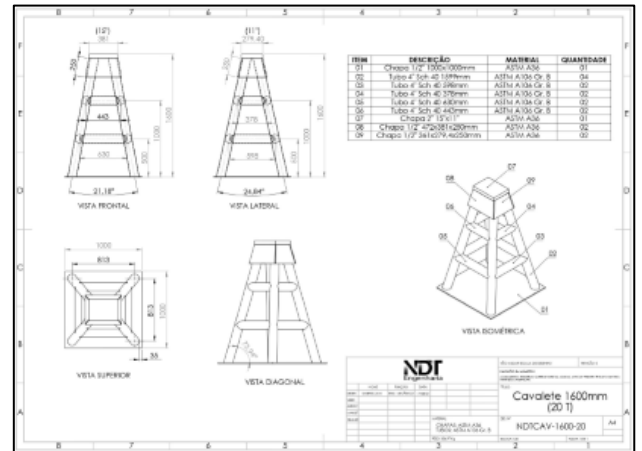
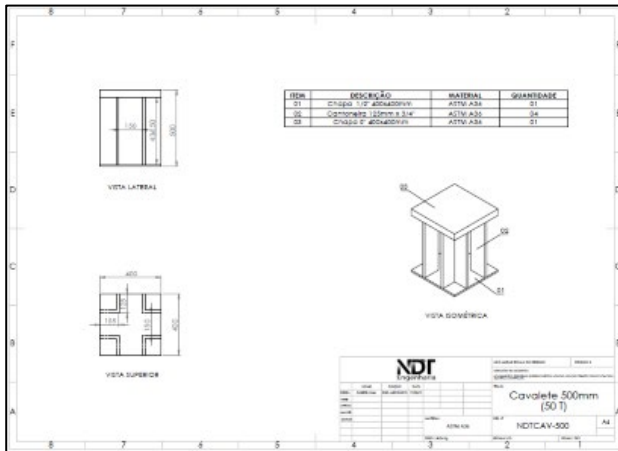
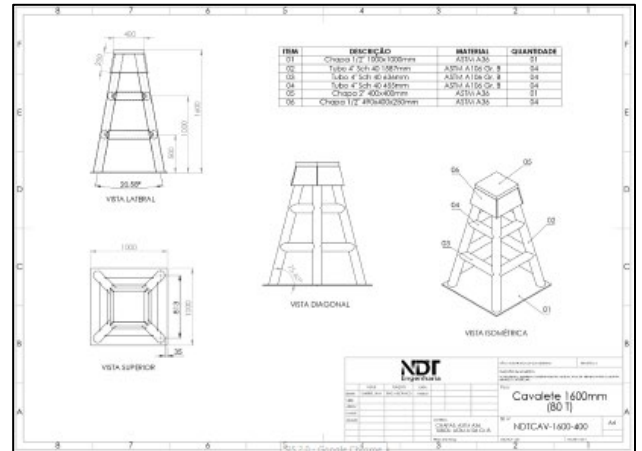
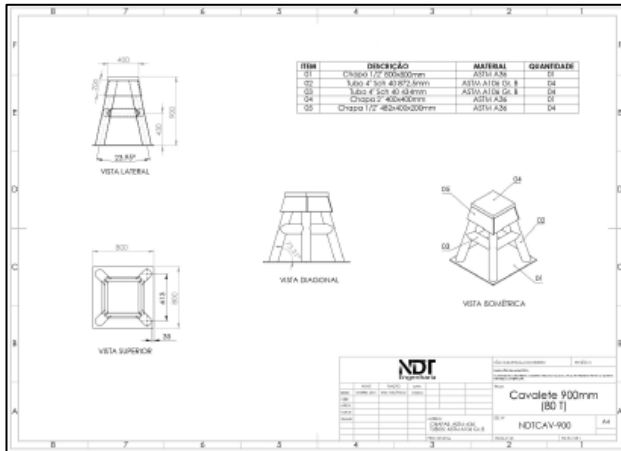


3.0 Implementation Steps

After defining the strategy for the assembly process, we defined the technical specifications and developed the supplier to manufacture the Chassis Stands Kits.

The design with the isometric view was validated by our Mining Engineering and the calculation memorial was performed by the supplier's engineering.

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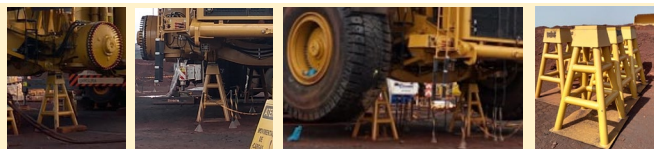
CHASSIS STANDS - 794AC

Jacking Locations (SEBU9099-08)

M0090261 - Assembly Procedure for the Off-Highway Truck - 794 AC

Position	Qty	Specification	Project	Illustration				
CLOSE THE GROUND - 900 MM								
Front of chassi	2	Height: 900 mm Upper Base: 2" x 400 x 400 mm Minimum capacity: 80 ton						
front Axle	1	Height: 500 mm Upper Base: 2" x 500 x 500 mm Minimum capacity: 50 ton						
Frame (Rear)	2	Height: 900 mm Upper Base: 2" x 400 x 400 mm Minimum capacity: 80 ton	Same project line 07					
Rear of the Machine	2	Height: 700 mm Upper Base: 2"x 300 x 300 mm Minimum capacity: 40 ton						
Superestructura	4	Height: 1600 mm Upper Base: 2" x 15" x 11" Minimum capacity: 20 Ton PN 455-8054 Note: Ensure the measurement of the upper base.		 - Used during initial machine build - Puts super structure at a workable height - Aids in installing hand rails on super structure Specifications - Dimensions: 838.2 x 838.2 x 1,524 mm (33 x 33 x 60 in) Reference - Stability Factor*: 7.5 - Weight: 443.5 kg (202 lb) - Working Load Limit: 12,000 kg (26,455 lb) - AS/NZS 2538:2004 <table><tr><th>Part Number</th><th>Description</th></tr><tr><td>455-8054</td><td>Super Structure Stand</td></tr></table>	Part Number	Description	455-8054	Super Structure Stand
Part Number	Description							
455-8054	Super Structure Stand							

TIRE AND RIM - INSTALL - (M0155539-02)

Front of the Machine	2	Height: 1600 mm Upper Base: 2" x 400 x 400 mm Minimum capacity: 80 ton		
Rear of the Machine	2	Height: 1600 mm Upper Base: 2" x 400 x 400 mm Minimum capacity: 80 ton	Same project line 13	

4.0 Benefits

- Standardization of Chassis Stands for assembly of **794AC** trucks.
- These Supports are manufactured according to technical specifications, with approved structural calculation.
- Chassis Stands with rubber top base to prevent metal-to-metal contact.
- Kits are available in the assembly area before equipment arrival.

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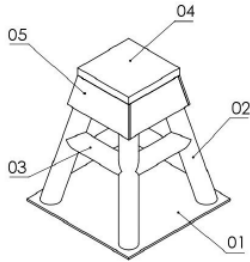
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- Cost reduction with cargo handling, as the main chassis was unloaded according to the layout and on the front axle.

5.0 Resources Required

900 mm

Capacity: 80 Metric Tons.

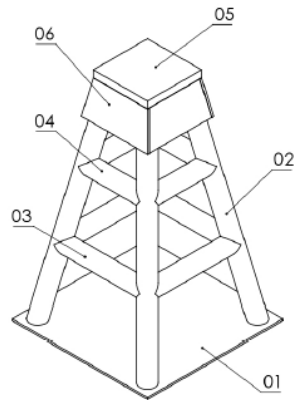


VISTA ISOMÉTRICA

ITEM	DESCRIPTION	MATERIAL	Qty
1	Plate 1/2" 800x800mm	ASTM A36	1
2	Tube 4" Sch 40 872,5mm	ASTM A106 Gr.B	4
3	Tube 4" Sch 40 434mm	ASTM A106 Gr.B	4
4	plate 2" 400x400mm	ASTM A36	1
5	Plate 1/2" 482x400x200mm	ASTM A36	4

1600 mm.

Capacity: 80 Metric Tons.

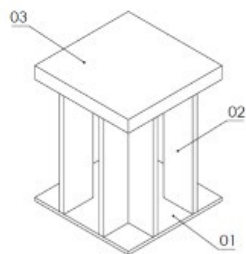


VISTA ISOMÉTRICA

ITEM	DESCRIPTION	MATERIAL	Qty
1	Plate 1/2" 1000x1000mm	ASTM A36	1
2	Tube 4" Sch 40 1587mm	ASTM A106 Gr.B	4
3	Tube 4" Sch 40 636mm	ASTM A106 Gr.B	4
4	Tube 4" Sch 40 455mm	ASTM A106 Gr.B	4
5	Plate 2" 400x400mm	ASTM A36	1
6	Plate 1/2" 490x400x250mm	ASTM A36	4

500 mm.

Capacity: 50 Metric Tons.



VISTA ISOMÉTRICA

ITEM	DESCRIPTION	MATERIAL	Qty
1	Plate 1/2" 400x400mm	ASTM A36	1
2	Angle bracket 125mm x 3/4"	ASTM A36	4
3	Plate 2" 400x400mm	ASTM A36	1

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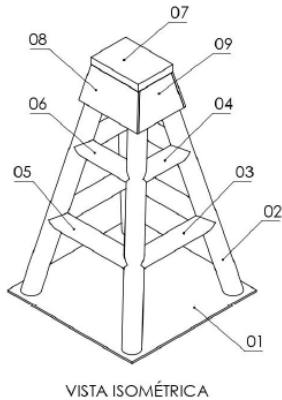
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1600 mm, (Superstructure)

Capacity: 20 Metric Tons



ITEM	DESCRIPTION	MATERIAL	Qty
1	Plate 1/2" 1000x1000mm	ASTM A36	1
2	Tube 4" Sch 40 1599mm	ASTM A106 Gr.B	4
3	Tube 4" Sch 40 598mm	ASTM A106 Gr.B	2
4	Tube 4" Sch 40 378mm	ASTM A106 Gr.B	2
5	Tube 4" Sch 40 630mm	ASTM A106 Gr.B	2
6	Tube 4" Sch 40 443mm	ASTM A106 Gr.B	2
7	Plate 2" 15"x11"	ASTM A36	1
8	Plate 1/2" 472x381x250mm	ASTM A36	2
9	Plate 1/2" 361x279,4x250mm	ASTM A36	2

5.1 Related costs.

The investment for manufacturing 1 set of stands in Brazil was approximately 50 thousand dollars.

All the stands meet the client's internal safety standards rules. The project was design by engineer and registered at the Regional Council of Engineering - CREA Brazil.

6.0 Supporting Attachments / References

794AC Service Manual.

Dealer & Product Group.

Customer Safety Requirements. **RAC's**. (Requisitos de atividades críticas).

NDT – Engineering.

Customer Feedback Survey.

Drawing attached for chassis stands

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

SOTREQ (Northern Brazil) would like to express its deep admiration for the professionals involved in improving the maintenance and repair processes developed by Mining Engineering.

SOTREQ would also like to thank the indispensable technical contribution of Doug Friedrich of Caterpillar Inc.

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